

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 13

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

BLANK PAGE

- 1 The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?
- A** -9
- B** 5
- C** -4
- D** -3
- E** 4
- F** -5

- 2 Find $\int \left(x + \frac{1}{x}\right)^2 dx$.
- A** $\frac{x^3}{3} + 2x - \frac{1}{x} + c$
- B** $\frac{3}{x^3} - \frac{1}{x} + c$
- C** $\frac{1}{3^3} \left(x + \frac{1}{x}\right)^3 + c$
- D** $\frac{x^3}{3} + 2x + \frac{1}{x} + c$
- E** $\frac{3}{x^3} + 2x + c$
- F** $x^3 + 2x - \frac{1}{x} + c$

3 Solve $4^{x+1} = 8^{2x-3}$.

A $x = \frac{11}{4}$

B $x = \frac{5}{4}$

C $x = \frac{9}{4}$

D $x = \frac{7}{2}$

E $x = 2$

F $x = 4$

- 4** A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?
- A** The argument is wrong because 2 is even and prime: being divisible by 2 does not stop 2 itself being prime.
 - B** The argument is correct.
 - C** The argument is wrong because some odd numbers are not prime.
 - D** The argument is wrong because 1 is odd and not prime.
 - E** The argument is wrong because 0 is even.

- 5 Consider the claim: **for every positive integer n , $n^2 + n$ is even.** Which of the following is a complete and valid proof?
- A** $n^2 + n$ is even because both n^2 and n are even.
- B** If n is even then n^2 is even and so $n^2 + n$ is even.
- C** Suppose $n^2 + n$ is odd. Then n is odd, so n^2 is odd, so $n^2 + n$ is even, which is a contradiction... unless n is even.
- D** $1^2 + 1 = 2$, $2^2 + 2 = 6$, $3^2 + 3 = 12$, which are all even, so the claim holds.
- E** $n^2 + n = n(n + 1)$ is the product of two consecutive integers, one of which must be even, so the product is even.

6 Here is a purported proof that $1 = 2$.

I Let $a = b$, with $a \neq 0$.

II Then $a^2 = ab$.

III So $a^2 - b^2 = ab - b^2$, i.e. $(a - b)(a + b) = b(a - b)$.

IV Dividing both sides by $(a - b)$: $a + b = b$.

V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

A IV

B II

C III

D V

E I

- 7 Let S be the statement: **every even number greater than 2 can be written as the sum of two primes.** A single counterexample to S would be
- A an odd number that is not the sum of two primes
 - B an even number greater than 2 that can be written as the sum of two primes in two different ways
 - C two primes whose sum is odd
 - D an even number that is the sum of two primes
 - E an even number greater than 2 that is not the sum of any two primes

- 8** Which of the following is the negation of '**there exists a positive integer n such that $n^2 + n + 41$ is not prime**'?
- A** For every positive integer n , $n^2 + n + 41$ is not prime.
 - B** There exists a positive integer n such that $n^2 + n + 41$ is prime.
 - C** For every positive integer n , $n^2 + n + 41$ is prime.
 - D** For some positive integer n , $n^2 + n + 41$ is prime.
 - E** There is no positive integer n for which $n^2 + n + 41$ is prime.

- 9 A proof that **if a and b are positive real numbers then $\frac{a+b}{2} \geq \sqrt{ab}$ ** is given with its lines scrambled.

I Hence $a + b \geq 2\sqrt{ab}$, and dividing by 2 gives the result.

II Expanding, $a - 2\sqrt{ab} + b \geq 0$.

III Since squares are non-negative, $(\sqrt{a} - \sqrt{b})^2 \geq 0$.

Which order gives a valid proof?

A III, II, I

B I, II, III

C II, III, I

D III, I, II

E II, I, III

10 Here is a proof that $\sqrt{2}$ is irrational, with its steps in the wrong order.

I Then $p^2 = 2q^2$, so p^2 is even, so p is even; write $p = 2k$.

II Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ where p and q are positive integers with no common factor.

III This contradicts the assumption that p and q have no common factor, so $\sqrt{2}$ is irrational.

IV Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, so q is even.

Which is the correct order?

A II, I, IV, III

B I, II, IV, III

C II, IV, I, III

D I, IV, II, III

E II, I, III, IV

- 11** *P*: 'a solid cylinder and a solid cube have equal volumes'. *Q*: 'the cylinder and the cube have equal surface areas'. Which one of the following is true?
- A** *P* is necessary and sufficient for *Q*.
 - B** *P* is necessary but not sufficient for *Q*.
 - C** *P* is sufficient but not necessary for *Q*.
 - D** *P* is neither necessary nor sufficient for *Q*.

12 An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

I If $d > 0$ then every term after some point is positive.

II The sum of the first n terms is a quadratic function of n .

III If a and d are integers then S_n is an integer for every n .

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 13** A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

- A** 16
- B** 20
- C** 14
- D** 8
- E** 17
- F** 12

14 n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I** There is exactly one such value of n .
- II** n must be odd.
- III** One of n , $n + 2$, $n + 4$ must be divisible by 3.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

15 Consider the statement: ****for all real x , $2^x > x^2$.**** Which of the following is a counterexample?

A $x = 5$

B $x = 3$

C $x = 0$

D $x = 10$

E $x = 1$

16 Claim: the sum of the first n odd numbers is n^2 .

I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.

II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.

III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.

IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

A The proof is completely correct.

B The first error is on line II: you cannot assume the claim.

C The first error is on line III: the next odd number is $2n - 1$.

D The first error is on line IV: it only shows the claim for $n + 1$.

E The first error is on line I.

17 n is a positive integer greater than 1. Which of the following statements is/are true?

I If n is not prime, then n has a prime factor at most $\sqrt{n}$.

II If n has no prime factor at most $\sqrt{n}$, then n is prime.

III If n is prime, then n has no factor between 1 and n exclusive.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 18** Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?
- A** Every friend shakes hands with exactly two others.
 - B** Every friend shakes hands with exactly four others.
 - C** The numbers of handshakes made are 1, 1, 2, 2, 2.
 - D** The numbers of handshakes made by the five friends are 4, 4, 4, 4, 1.
 - E** Exactly one friend shakes no hands at all.

19 The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

I $f(x) > 0$ for all real x

II The graph of $y = f(x)$ is symmetric about the line $x = 3$

III The equation $f(x) = 4$ has exactly one real solution

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

20 n is a positive integer. Which of the following statements is/are true?

I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.

II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.

III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST